

Thesis Defense:
Invariant measures for the stochastic heat equation with
no drift term

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March 29, 2022

The stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x), & t \geq 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The Stochastic Heat Equation is interpreted as follows:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(u(s, y))W(ds, dy)$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

- A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

The noise term $\dot{W}$

- $W = \{W(\phi) : \phi \in \mathcal{S}(\mathbb{R}^d)\}$ is a centered Gaussian process uniquely defined through the following covariance functional:

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$. We have the following nomenclature.

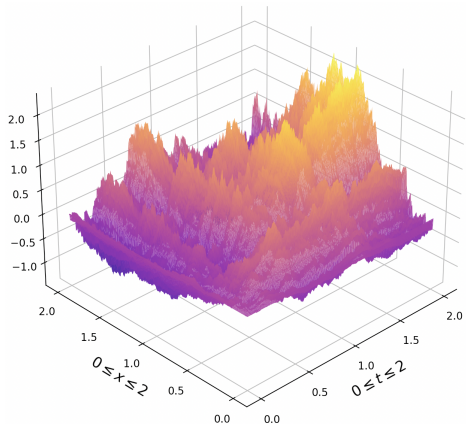
- *Space-time white noise*: $\Gamma(dx) = \delta_0(dx)$.
- *Time white and spatially colored*: $\Gamma(dx) = f(x)dx$ where f is a non-negative and non-negative definite function referred to as the *Correlation function* and its Fourier transform,

$$\hat{f}(\xi) := \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-ix \cdot \xi) f(x)dx,$$

is referred to as the *spectral density*.

Proposition (DaPrato and Zabczyk [7, Propsition 4.19])

A space-time white noise, dB , is the distributional derivative of a Brownian sheet, $B(t, x)$.



White noise expansion DaPrato [7]

If $t \geq 0$ and $0 \leq x \leq \pi$ then,

$$B(t, x) = \sum_{n=0}^{\infty} \frac{\beta_n(t)}{n + \frac{1}{2}} \sqrt{\frac{2}{\pi}} \sin \left[\left(n + \frac{1}{2} \right) x \right]$$

Walsh [9]

$$\int_t^{t+\delta_t} \int_x^{x+\delta_x} dB(ds, dy) \sim N(0, \delta_t \delta_x)$$

Brownian Sheet Simulation (Walsh [9]):

$$B(t, x) \sim B(t - \delta_t, x) + B(t, x - \delta_x) - B(t - \delta_t, x - \delta_x) + N(0, \delta_t \delta_x)$$

$$B(t, 0) = B(0, x) = 0 \text{ for all } t, x > 0.$$

Existence and uniqueness with space-time white noise

- A necessary condition for global existence, *Dalang's condition* [6], is the following:

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) < \infty, \quad \forall t > 0.$$

- Direct calculations yields that,

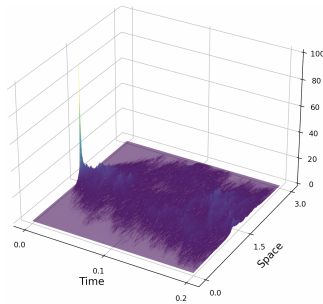
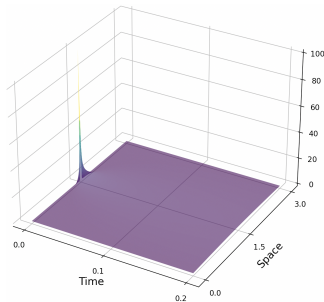
$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) = \frac{\Gamma(d/2)}{2(2\pi)^d} \int_0^t \frac{1}{s^{d/2}} ds,$$

which forces,

$$d/2 < 1 \quad \Longleftrightarrow \quad d < 2.$$

- Thus a global space-time white noise solution may only exist when the spatial dimension is 1.

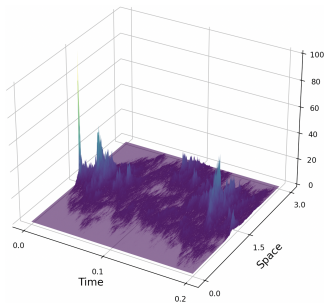
$$u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W} \text{ with } u_0(x) = \delta_0(x) \quad (\text{white noise})$$



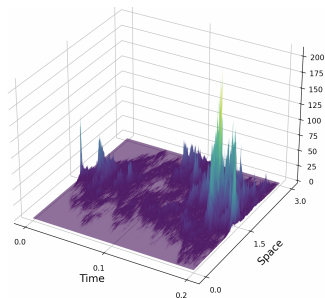
- Simulations done by using a finite difference scheme [5].
- We partition (space) $[0, 3]$ into n_x pieces of size $\delta_x = 3/n_x$.
- We partition (time) $[0, .2]$ into n_t pieces of size $\delta_t = .2/n_t$.
- To control error, we assume $n_x^2 \delta_t < \frac{1}{2}$ which implies $n_t \gg n_x$.

$$u(t + \delta_t, x) = u(t, x) + n_x^2 \delta_t (u(t, x + \delta_x) + u(t, x - \delta_x) - 2u(t, x)) \\ + n_x \lambda u(t, x) \text{normal}(0, \delta_t \delta_x).$$

Intermittency of $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ (white noise)



$\lambda = 1.8$



$\lambda = 2$

● Notice the enlargement of the peaks as λ increases. We have the following theorem.

Theorem (Chen and Dalang [1, Theorem 2.4])

The second moments of the stochastic heat equation driven by space-time white noise grow exponentially in time.

Will the second moments ever be bounded in time?

- Several slides below will require the second moments to be bounded in time:

$$\sup_{t \geq 0} \|u(t, x)\|_2 < \infty, \quad \text{where } \|X\|_p = [\mathbb{E}(|X|^p)]^{1/p}.$$

- This is possible, however this will require us to only consider $d \geq 3$, and therefore we must only consider a spatially colored noise.
- Recall from above, that the covariance of a spatially colored noise is given by the following:

$$\begin{aligned} \mathbb{E}[W(\phi)W(\psi)] &= \int_0^\infty ds \int_{\mathbb{R}^d} dx f(x)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) f(x - y) \psi(s, y) \\ &= (2\pi)^{-d} \int_0^\infty ds \int_{\mathbb{R}^d} d\xi \hat{f}(\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)} \end{aligned}$$

$$u_t - \frac{1}{2} u_{xx} = b(u) \dot{W} \text{ with } u(0, \cdot) = \mu(\cdot) \quad (\text{colored noise})$$

Theorem (Chen and Huang [2, Theorem 1.2])

There exists a unique global mild solution under the following:

- $b : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a $L_b, L_0 > 0$ so for all $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0.$$

- The initial measure, $\mu(dx)$ (resp. $\varphi(x)dx$), satisfies for some $a > 0$,

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty \quad \left(\text{resp. } \int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty \right),$$

where $|\mu| = \mu^+ + \mu^-$ is the *Hahn decomposition*.

- The spectral density, $\hat{f}$, satisfies *Dalang's condition* for some $\beta > 0$:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

Bounded moments in time

Theorem (Chen and Kim [4, Theorem 1.3 and Lemma 2.5])

Suppose that we strengthen Dalang's condition and the Lipschitz assumption to the following:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Then the following holds:

$$\begin{aligned} \|u(t, x)\|_2 &\leq C_2 \left[1 + \left(G(t, \cdot) * |\mu| \right)(x) \right] \\ &= C_2 \left[1 + \int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy) \right], \quad \text{for all } t \geq 0. \end{aligned}$$

- If $\mu = \delta_0$ then $\|u(t, x)\|_2 \leq C_2 [1 + (2\pi t)^{-d/2}]$.
- Later, it will suffice consider μ such that $\sup_{t \geq t_0} \|u(t, x)\|_2 < \infty$.

Transition functions and their semigroup

Definition (Markovian Transition Functions)

- Let H be a Hilbert space and $\mathcal{B}(H)$ be the borel sets of H .
- $\{P_t(x, \Gamma) : t \geq 0, x \in H, \Gamma \in \mathcal{B}(H)\}$ forms a family of *Markovian transition functions* if,

(1) $x \mapsto P_t(x, \Gamma)$ if measurable,

(2) $\Gamma \mapsto P_t(x, \Gamma)$ is a probability measure on H ,

(3) $P_{t+s}(x, \Gamma) = \int_H P_t(x, dy) P_s(y, \Gamma)$,

(4) $P_0(x, \Gamma) = \mathbf{1}_\Gamma(x)$.

Definition (Transition Semigroup)

The above transition functions form the following semi group on $\mathcal{B}(H)$, which denotes the bounded Borel measurable functions on H :

$$P_t \varphi(x) = \int_H \varphi(y) P_t(x, dy), \quad \varphi \in \mathcal{B}(H).$$

Invariant measures

Definition (Invariant measures)

- A probability measure, η , on H is said to be *invariant* for the transition functions P_t if for all $\Gamma \in \mathcal{B}(H)$ and $t > 0$,

$$P_t^*(\eta)(\Gamma) := \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_0(x, \Gamma) \eta(dx) = \int_H \mathbf{1}_\Gamma(x) \eta(dx) = \eta(\Gamma),$$

which is equivalent to,

$$\int_H P_t \varphi(x) \eta(dx) = \int_H \varphi(x) \eta(dx), \quad \text{for all } \varphi \in \mathcal{B}(H).$$

- This implies that the η -average of $x \mapsto P_t(x, \Gamma)$ is $\eta(\Gamma)$ and it is independent of time:

$$\text{AVG}_\eta[P_t(\cdot, \Gamma)] = \frac{1}{\eta(H)} \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_t(x, \Gamma) \eta(dx) = \eta(\Gamma).$$

Invariant measures for the stochastic heat equation

- First we define the law of the solution as follows:

$$\mathcal{L}(u(t, \cdot; \zeta))(\Gamma) = \mathbb{P}[\omega \in \Omega : u(t, \cdot; \zeta) \in \Gamma], \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H),$$

and for the mean time we are intentionally vague about H .

- Our intention is for

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma)$$

to form a Markovian semigroup.

- Established theory by DaPrato and Zabczyk [7] requires the solution,

$$\zeta \mapsto u(t, \cdot; \zeta),$$

to be almost surely H -valued for $t > 0$ and $\zeta \in H$.

- So what is H ?

$H = L^2_\rho(\mathbb{R}^d)$: weighted square integrable functions

Definition (Admissible Weight)

A positive, bounded and continuous $L^1(\mathbb{R}^d)$ function, ρ , is an *admissible weight* if

$$\left(G(t, \cdot) * \rho(\cdot) \right)(x) \leq C_\rho \rho(x), \quad \text{for all } t \in [0, T].$$

Definition ($L^2_\rho(\mathbb{R}^d)$)

The space $H := L^2_\rho(\mathbb{R}^d)$ is the space of ρ -weighted square integrable functions:

$$\|\zeta\|_\rho^2 := \int_{\mathbb{R}^d} |\zeta(x)|^2 \rho(x) dx < \infty.$$

Moreover, H is a Hilbert space with inner product defined as

$$\langle \zeta, \psi \rangle_\rho := \int_{\mathbb{R}^d} \zeta(x) \psi(x) \rho(x) dx.$$

Weight examples

- The following two are examples of admissible weights:

$$\exp(-a|x|), \text{ for } a > 0 \quad \text{and} \quad \frac{1}{1 + |x|^a}, \text{ for } a > d.$$

- One can show that the following is not admissible for any $a > 0$,

$$\exp(-a|x|^2),$$

- Numerically it is suggested that for any $1 < b < 2$ and $a > 0$ that

$$\exp(-a|x|^b)$$

may not be admissible, but this has not been proven.

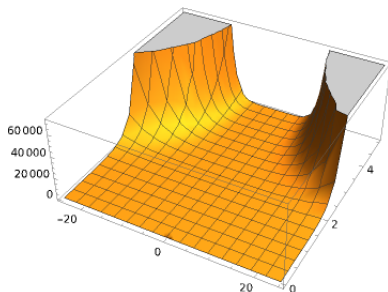


Figure: Plot of $\frac{(G(t,\cdot)*\exp(-|\cdot|^{1.25}))(x)}{\exp(-|x|^{1.25})}$

The solution is almost surely H -valued.

Theorem

Let ρ be an admissible weight and suppose that

$$\Upsilon(0) = \int_{\mathbb{R}^d} \frac{\widehat{f}(\xi)}{|\xi|^2} d\xi < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Now suppose our initial measure, μ , satisfies the following:

$$\mathcal{G}_\rho(t; \mu) := \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y)(1 + |\mu|)(dy) \right)^2 \rho(x) dx < \infty.$$

Then for all $t > 0$,

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; \mu) < \infty.$$

In particular, for any $\zeta \in H$ and for all $T > 0$, there exists a C_T such that

$$\mathcal{G}_\rho(t; \zeta) \leq 2 \|\rho\|_{L^1(\mathbb{R}^d)} + 2C_T \|\zeta\|_\rho < \infty, \quad \text{for all } t \in [0, T].$$

Invariant measures for the stochastic heat equation

- So the solution acts as an operator, mapping H to H .
- Moreover, the Markov property of the solution [7, Th 9.21] implies that

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma), \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H)$$

for a family of Markovian transition functions.

Definition

A measure η is invariant for the stochastic heat equation if η is invariant for the transition functions P_t . In other words,

$$P_t^*(\eta)(\Gamma) = \int_H P_t(\zeta, \Gamma) \eta(d\zeta) = \eta(\Gamma), \quad \text{for all } \Gamma \in \mathcal{B}(H).$$

Convergence of measures

- Let $\mathcal{M}(H)$ and $C_b(H)$ denote the probability measures and bounded continuous functions on H .
- For $\varphi \in C_b(H)$ and $\nu \in \mathcal{M}(H)$,

$$\langle \varphi, \nu \rangle = \int_H \varphi(\zeta) \nu(d\zeta).$$

Definition

We say ν_n converges *weakly* to η in $\mathcal{M}(H)$ if

$$\langle \varphi, \nu_n \rangle \rightarrow \langle \varphi, \nu \rangle, \quad \text{as } n \rightarrow \infty.$$

We will use the notation, $\nu_n \rightarrow \nu$, to denote the weak convergence of measures.

Some existence results

Theorem (DaPrato [7, Propositions 11.2 and 11.3])

If $\zeta_1, \zeta_2 \in H$ and $\eta_1, \eta_2 \in \mathcal{M}(H)$ are such that

$$\mathcal{L}\left(u(t, \cdot; \zeta_1)\right) \rightarrow \eta_1 \text{ as } t \rightarrow \infty \quad \text{or} \quad \frac{1}{t_n} \int_0^{t_n} \mathcal{L}\left(u(s, \cdot; \zeta_2)\right) ds \rightarrow \eta_2 \text{ as } n \rightarrow \infty,$$

for some $t_n \uparrow \infty$, then both η_1 and η_2 are invariant.

Theorem (DaPrato [7, Proposition 11.5])

Suppose η is invariant for the stochastic heat equation and $\zeta \in H$ is random and $\mathcal{L}(\zeta) = \eta$. Then $\mathcal{L}\left(u(t, \cdot; \zeta)\right)$ is stationary:

$$\mathbb{E}\left[\varphi_1(u(t_1 + h, \cdot; \zeta)) \cdots \varphi_n(u(t_n + h, \cdot; \zeta))\right]$$

is independent of h for any $n \in \mathbb{N}$ and $\varphi_i \in C_b(H)$.

Under what setup has an invariant measure been proven to exist?

- Research on invariant measures for $u_t - \frac{1}{2}\Delta u = f(u) + b(u)\dot{W}$ on the whole space with a globally Lipschitz b is not vast.
- Tessitore and Zabczyk [8] is the only other paper which studies $u_t - \frac{1}{2}\Delta u = b(u)\dot{W}$ on the whole space with a globally Lipschitz b .
- Tessitore and Zabczyk showed there exists an $\eta \in \mathcal{M}(H)$ such that,

$$\frac{1}{t_n} \int_1^{t_n} \mathcal{L}\left(u(s, \cdot; 1)\right) ds \rightarrow \eta, \quad \text{as } n \rightarrow \infty,$$

under the following assumption on the spectral density $\widehat{f}$:

$$\int_{\mathbb{R}^d} |\xi|^{2-d} \left[\left| \mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \right| * \left| \mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \right| \right](\xi) d\xi < \infty.$$

- If one could find a f such that $\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \geq 0$ and $\widehat{f} \geq 0$ then

$$\begin{aligned}
 & \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| * \left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| \right] (\xi) \, d\xi \\
 &= \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) * \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right] (\xi) \, d\xi \\
 &= K \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\mathcal{F}^{-1} \left(\widehat{f} \right) \right] (\xi) \, d\xi \\
 &= K \int_{\mathbb{R}^d} |\xi|^{2-d} f(\xi) \, d\xi \\
 &= K \int_{\mathbb{R}^d} |\xi|^{-2} \widehat{f}(\xi) \, d\xi,
 \end{aligned}$$

Which is precisely Dalang's condition.

- $\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \geq 0$ is far too restrictive and no such examples of f have been given.

- $\mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \geq 0$ requires $\sqrt{\widehat{f}}$ to be non-negative definite and this may not be true even if $\widehat{f}$ is non-negative and non-negative definite.

Example

Consider the pair

$$f(x) = (2\pi)^{-1} \sin(x^2)/x^2 \quad \text{and} \quad \widehat{f}(\xi) = \frac{1}{4} \max\{2 - |\xi|, 0\}.$$

Both f and $\widehat{f}$ is clearly non-negative and thus both are non-negative definite. However, direct calculation shows that

$$\mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right)(x) = \frac{\sqrt{\frac{\pi}{2}} \left(-\cos(2x) \mathcal{S}\left(\frac{2\sqrt{x}}{\sqrt{\pi}}\right) + \sin(2x) \mathcal{C}\left(\frac{2\sqrt{x}}{\sqrt{\pi}}\right) \right)}{x^{3/2}}$$

Where $\mathcal{S}$ and $\mathcal{C}$ are the sine and cosine Fresnel integrals,

$$\mathcal{C}(z) = \int_0^z \cos\left(\frac{\pi t^2}{2}\right) dt \quad \text{and} \quad \mathcal{S}(z) = \int_0^z \sin\left(\frac{\pi t^2}{2}\right) dt.$$

$\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$ takes positive and negative values

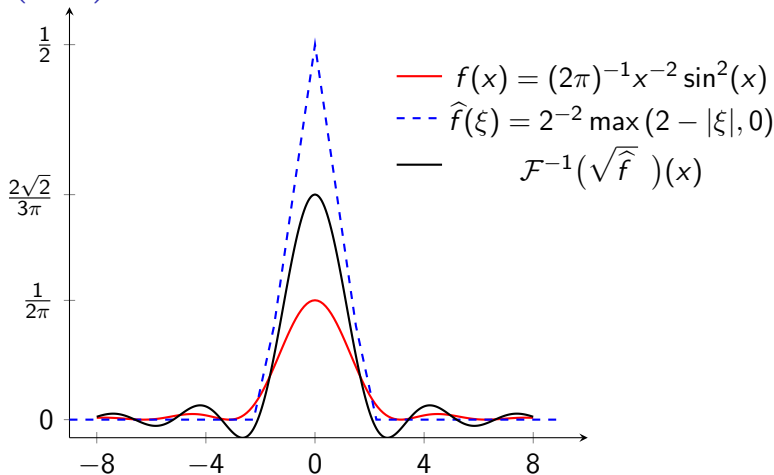


Figure: One example that $\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$ assumes both positive and negative values.

- A simpler condition needs to be introduced in order to give a full example.
- Recall that

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_{\rho} \right) \leq C \mathcal{G}_{\rho}(t; \mu) := C \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y)(1 + |\mu|)(dy) \right)^2 \rho(x) dx$$

Theorem

Suppose that the initial measure, μ , the diffusion term, b , and the spectral density, $\hat{f}$ satisfy

$$\Upsilon_{\alpha}(0) := \int_{\mathbb{R}^d} \frac{f(\xi) d\xi}{|\xi|^{2(1-\alpha)}} < \infty, \quad \limsup_{t>0} \mathcal{G}_{\rho}(t; \mu) < \infty, \quad \text{and} \quad L_b^2 < \frac{1}{2^7 \Upsilon(0)},$$

for some $\alpha \in (2^7 \Upsilon(0) L_b^2, 1)$ and also for $\alpha = 0$. Then there exists an invariant measure for the stochastic heat equation. More precisely, the following weak limit exists for each $t_0 > 0$:

$$\frac{1}{t_0} \int_{t_0}^{t_n+t_0} \mathcal{L} \left(u(s, \cdot; \mu) \right) ds \rightarrow \eta_{t_0},$$

where η_{t_0} is the invariant measure.

- We will now give some permissible initial conditions and spectral densities.

Permissible spectral density

- Recall that $\widehat{f}$ must satisfy both $\alpha = 0$ and $\alpha \in (2^7 \Upsilon(0) L_b^2, 1)$:

$$\Upsilon_\alpha(0) := \int_{\mathbb{R}^d} \frac{f(\xi) \, d\xi}{|\xi|^{2(1-\alpha)}} < \infty.$$

- Recall that $d > 3$ and so $\widehat{f}$ may have slow blow up at the origin but must have fast decay at $\pm\infty$ to counter act the thick tails of $|\xi|^{2(1-\alpha)}$.
- This excludes the Riesz kernel.
- However, the Bessel kernel has polynomial blowup at the origin and exponential decay at $\pm\infty$ and will be permissible.

Bessel Kernel

- Let f_s denote the Bessel kernel. When $0 < s < d$ we have that

$$f_s(x) \sim \exp(-|x|/2) \text{ for } |x| > 2 \quad f_s(x) \sim \frac{1}{|x|^{d-s+2}} \text{ for } |x| < 2.$$

- The Fourier transform $\hat{f}$ is given by $\hat{f}(\xi) = (1 + |\xi|^2)^{s/2}$.

Example

When considering the Bessel kernel, direct calculation yields that for any $\alpha \in [0, 1]$ and $s > d - 2(1 - \alpha) > 0$ and $d \geq 3$,

$$\Upsilon_\alpha(0) = \frac{\Gamma\left(\frac{d}{2} - 1 + \alpha\right) \Gamma\left(\frac{s-d}{2} + 1 - \alpha\right)}{2^d \pi^{d/2} \Gamma(s/2)},$$

and when $\alpha = 0$, this simplifies to

$$\Upsilon(0) = \frac{\Gamma\left(\frac{2+s-d}{2}\right)}{2^{d-1} \pi^{d/2} (d-2) \Gamma(s/2)}.$$

Permissible initial conditions

(1) For any $\zeta \in L^\infty(\mathbb{R}^d)$ then for all $t \geq 0$

$$\mathcal{G}_\rho(t; \mu) \leq \left(1 + \|\zeta\|_{L^\infty(\mathbb{R}^d)}\right)^2 \|\rho\|_{L^1(\mathbb{R}^d)} < \infty.$$

(2) For $\delta_0(\cdot)$ and for all $t > 0$,

$$\mathcal{G}_\rho(t; \delta_0) \leq (2 + G(t, 0)^2) \|\rho\|_{L^1(\mathbb{R}^d)} < \infty.$$

(3) Consider $|\cdot|^{-\lambda}$ for any $\lambda \in (0, d)$. Then for all $t > 0$,

$$\mathcal{G}_\rho(t; |\cdot|^{-\lambda}) \leq 2(1 + Ct^{-\lambda}) \|\rho\|_{L^1(\mathbb{R}^d)} < \infty, \quad C = 2^{-\lambda} \left(\frac{\Gamma((d-\lambda)/2)}{\Gamma(d/2)} \right)^2,$$

- $\delta_0(\cdot) \notin L^2_\rho(\mathbb{R}^d)$ but still permissible.
- $|\cdot|^{-\lambda} \notin L^2_\rho(\mathbb{R}^d)$ for $\lambda \in (d/2, d)$, however, it is still applicable.

References



L. Chen and R. Dalang (2015)

Moments and growth indices for the nonlinear stochastic heat equation with rough initial conditions

The Annals of Probability 43, 6, 3006–3051



L. Chen and J. Huang

Comparison principle for stochastic heat equation on $\mathbb{R}^d$

Ann. Probab., 47.2 (2019) 989–1035.



L. Chen, Y. Hu, D. Nualart (2019)

Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$.

Stochastic Processes and their Applications, 129 5073–5112.



L. Chen and K. Kim (2019)

Nonlinear stochastic heat equation driven by a spatially colored noise: moments and intermittency.

Acta Mathematica Scientia 39B(3), 645–668



A. Davie and J. Gaines (2000)

Convergence of numerical schemes for the solution of parabolic stochastic